

Advanced (Carolina Reaper)

Q1. Solve $|x| = 5$

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. Solve $|2x| = 8$

- (A) $x = 2$ or $x = -4$
- (B) $x = 4$ or $x = -2$
- (C) $x = 1$ or $x = -4$
- (D) $x = 4$ or $x = -4$
- (E) $x = 2$ or $x = -2$

Q3. Solve $|x - 2| = 3$

- (A) $x = 5$ or $x = -1$
- (B) $x = 2$ or $x = -3$
- (C) $x = 1$ or $x = 4$
- (D) $x = 3$ or $x = 2$
- (E) $x = 0$ or $x = 5$

Q4. Solve $|x + 1| = 2$

- (A) $x = 3$ or $x = -1$
- (B) $x = 1$ or $x = -3$
- (C) $x = 2$ or $x = -4$
- (D) $x = 0$ or $x = -2$
- (E) $x = 1$ or $x = -2$

Q5. Solve $|3x| = 12$

- (A) $x = 2$ or $x = -4$
- (B) $x = 4$ or $x = -4$
- (C) $x = 3$ or $x = -4$
- (D) $x = 4$ or $x = -2$
- (E) $x = 4$ or $x = -1$

Q6. Solve $|2x - 1| = 5$

- (A) $x = 2$ or $x = -3$
- (B) $x = 3$ or $x = -2$
- (C) $x = 1$ or $x = 4$
- (D) $x = 3$ or $x = -1$
- (E) $x = 2$ or $x = 1$

Q7. Solve $|x + 2| = 4$

- (A) $x = 6$ or $x = -2$
- (B) $x = 2$ or $x = -6$
- (C) $x = 1$ or $x = 5$
- (D) $x = 0$ or $x = -4$
- (E) $x = 2$ or $x = 1$

Q8. Solve $|x - 3| = 2$

- (A) $x = 1$ or $x = 5$
- (B) $x = 3$ or $x = 2$
- (C) $x = 5$ or $x = 0$
- (D) $x = 2$ or $x = 4$
- (E) $x = 1$ or $x = 4$

Open Ended Questions (FRQ)

Q9. Solve the equation $|2x + 1| = |x - 2|$ and check for extraneous solutions

Q10. Solve the equation $|x| = |2x - 3|$ and check for extraneous solutions

Chef's Tips

- Tip 1: Check for extraneous solutions
- Tip 2: Use the definition of absolute value
- Tip 3: Be careful with negative signs